



iJRASET

International Journal For Research in
Applied Science and Engineering Technology



INTERNATIONAL JOURNAL FOR RESEARCH

IN APPLIED SCIENCE & ENGINEERING TECHNOLOGY

Volume: 13 Issue: V Month of publication: May 2025

DOI: <https://doi.org/10.22214/ijraset.2025.70716>

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Android Based Blood Donation Management System

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Abstract: *Android technology revolutionizes hospital operations by streamlining electronic health records, patient monitoring, enhancement and appointment scheduling. Android apps drive efficiency and enhancements in patient care. Proposed System also help in monitoring the blood donations and manage the blood donors and requesters using Android app. The existing system propose a private Ethereum blockchain based solution to automate blood donation management. It stores non-critical and large data off-chain using decentralized storage of InterPlanetary File System (IPFS). Existing system have several critical issues like lack of authenticity, struggles with the tracking fake donors and requestors. Also lack of real time data entry. I am proposing a solution in which I have implemented OTP based verification system which improves the authenticity of system, also OCR based technology that is implemented in proposed system to improve the tracking of fake donors and requestors. Firebase database system is implemented to provide and improve real time data entry.*

Keywords: *Android technology; Hospital operations; Electronic health records; Patient monitoring; Appointment scheduling; Blood donation management; Blood donors; Requesters; Private Ethereum blockchain; Decentralized storage; InterPlanetary File System (IPFS); Authenticity; Fake donors; Fake requesters; Real-time data entry; OTP-based verification; OCR technology; Firebase database.*

I. INTRODUCTION

In times of medical emergencies, the timely availability of blood can be a matter of life and death. The Blood Donor Management System (BDMS) is a groundbreaking initiative that bridges the gap between voluntary blood donors and those in desperate need of blood, ensuring that help is just a tap away. This innovative mobile application is designed to empower users by allowing seekers to identify potential donors in their vicinity, facilitating a swift response to urgent requests. Recognizing the current challenges in blood donation systems—such as inadequate authentication mechanisms and the risk of fraudulent activities—the BDMS incorporates an OTP-based authentication system to enhance user verification. Furthermore, to combat the issue of fraudulent requests, the system integrates an OCR-based verification process that cross-references donor and request details against hospital-issued tickets. In an era where efficiency and real-time data management are crucial, the BDMS utilizes Firebase Database to streamline user management and improve data handling processes. By harnessing technology, the Blood Donor Management System not only simplifies the donation process but also aims to create a more connected and responsive community dedicated to saving lives.

II. LITERATURE OVERVIEW

The management of blood donation systems has evolved significantly, with recent advancements focusing on automation, security, and efficiency. This literature review examines several notable works that explore innovative approaches to blood donor management, highlighting their contributions and limitations.

D. Hawashin, D. Amin, K. Salah, et al., “BLOCKCHAIN-BASED MANAGEMENT OF BLOOD DONATION,” 2021. This study proposes a private ethereum blockchain solution to automate blood donation management. The use of blockchain technology ensures enhanced security, traceability, and privacy throughout the donation process. The authors emphasize the role of the Interplanetary File System (IPFS) for decentralized storage, addressing the storage limitations commonly faced by traditional systems. The conclusion drawn from this research highlights that while blockchain enhances transparency and security, careful integration into healthcare systems is essential to address the complexities involved in implementation[1].

Y. Bhole, S. Patil, A. Ali, and H. Khan, “Blood Bank and Donor Management System,” 2023. This paper presents a centralized web-based system that automates blood donation management, facilitating real-time donor matching and tracking of blood supplies. The study points out that the centralized nature of the system simplifies donor management and offers a real-time search

functionality, enabling users to locate nearby donors quickly. However, the authors note that despite the automation improving efficiency and data accuracy, the system lacks real-time traceability, which could limit its effectiveness in urgent situations[2].

A. Choudhary, A. Bhardwaj, D. Devesh, and Prof. B. Bhushan, “Comprehensive Blood Bank Management System,” 2024. This work introduces a comprehensive solution that encompasses donor management, blood testing, transfusion, and reporting while ensuring regulatory compliance. The integration of regulatory standards guarantees safety and standardization in the blood donation process. Their findings indicate that full system automation, from donor registration to transfusion, significantly enhances operational efficiency. However, they stress the importance of robust data security measures to protect sensitive donor information, highlighting a crucial area that requires further attention[3].

Y. Bhole, S. Patil, et al., “A Research Paper on Blood Bank and Donor Management System,” 2023. In this study, the authors discuss a centralized system designed to automate the blood donation process, addressing existing inefficiencies. By streamlining data access and reducing manual intervention, the proposed system enhances the accuracy of donor registrations. The research concludes that full-system integration is essential for managing the complexities of blood bank operations, yet emphasizes the necessity of implementing proper security controls to safeguard user data[4].

V. Satpute, S. Patil, S. Taral, et al., “Blood Bank Management System,” 2023. This paper focuses on automating blood bank operations through a database management system (DBMS) and Java-based solution. Their approach aims to improve inventory and donor management by reducing manual errors and providing real-time data on blood stocks, which can help minimize wastage. While the study acknowledges the benefits of centralized systems for improved accessibility, it also raises concerns about the scalability of these systems to meet real-time demands, suggesting that modern cloud solutions may be necessary[5].

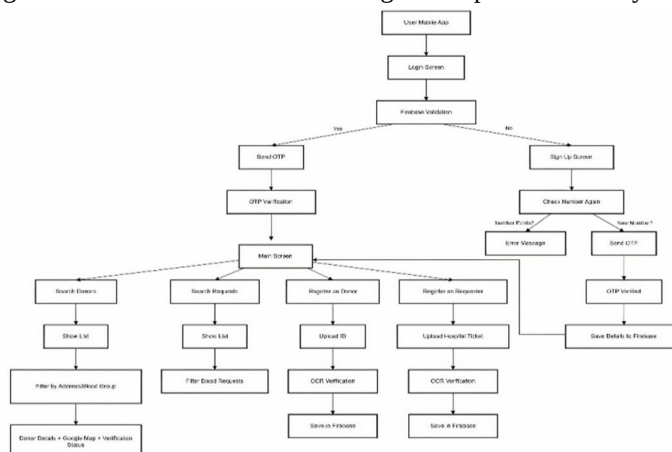
III. PROPOSED WORK

Blood Donation Management System is a platform that connects voluntary blood donors with individuals in need of blood during emergencies. The purpose of this research is to develop a mobile application that helps seekers identify blood donors near their location. Both donors and seekers must register by providing their basic details in the mobile application.

When there is a need, the seeker can raise a request through the application, which will be visible to all donors. Seekers can also locate nearby blood banks. Once a request is made, a notification is sent to all registered donors, allowing them to accept or decline the request. If a donor agrees to donate blood, their details will be shared with the seeker, who will also receive directions to the donor’s location. To ensure data security, the donor’s personal information is not shared without their consent. After the donation process, donors can update their status. Additionally, a one-time password (OTP) is generated to authenticate users during the registration process.

Current blood donor management systems face several challenges. One major issue is the lack of authentication mechanisms, leading to potential misuse due to the inability to verify user identities. To address this, an OTP-based authentication system is proposed to enhance user verification. Additionally, existing systems struggle to detect fraudulent donors and requests due to inadequate tracking and verification processes. Integrating an OCR-based verification system, which validates donor and request details directly from hospital-issued tickets, can significantly improve fraud detection.

Lastly, many systems suffer from inefficiencies in daily user management due to inadequate data handling. Implementing a Firebase Database for real-time data management can streamline user handling and improve overall system efficiency.



The proposed system architecture for the Blood Donation Management App is designed to provide a secure, efficient, and user-friendly experience. It begins with OTP-based login or signup, ensuring only verified users access the system. Firebase handles real-time validation and data storage. After successful login, users reach the main screen with options to search for donors or requesters, and to register themselves. The integration of OCR technology verifies donor IDs and hospital documents for authenticity, while Google Maps helps users find nearby matches quickly. This structured flow ensures reliability, transparency, and faster emergency response.

IV. IMPLIMENTATION DETAILS

A. Login Screen

The Login Screen allows users to securely log in using their mobile phone numbers. Upon entering the number, the app checks if it exists in the system. If it does, an OTP is sent for verification. Once verified, users are granted access to the main dashboard. The screen is simple, user-friendly, and ensures secure entry into the app.

B. Signup Screen

The Signup Screen is designed for new users whose phone numbers are not found in the database. Users fill out their basic details like name, age, blood group, and address. Before proceeding, the app checks if the number already exists to prevent duplication. After successful OTP verification, the user's data is saved to Firebase and access is granted to the main screen.

C. Main Dashboard Screen

The Main Dashboard displays user details at the top and provides four main options: Search Donors, Search Requesters, Register as Donor, and Request for Blood. The design is clean and intuitive, making it easy for users to quickly find what they need.

D. Search Donor Screen

This screen allows users to search for registered blood donors. Filters like blood group and address help narrow down search results. Upon selecting a donor, users can view detailed information including the donor's verification status and exact location shown on Google Maps for easier navigation.

E. Search Requester Screen

Similar to the donor search, this screen helps donors find individuals currently requesting blood. Users can search by blood group and hospital address. Detailed profiles and map locations ensure donors can locate and assist requesters effectively.

F. Register As Donor Screen

Donors can register themselves by providing their personal and medical details. To prevent fake entries, donors must upload a valid ID proof, which is verified using OCR technology. Only verified donors are listed publicly to ensure trust and reliability.

G. Request Blood Screen

Requesters can post their blood requirements by filling in necessary details like hospital name, required units, and blood group. They must upload a hospital ticket or medical document, which is validated using OCR. Verified requests appear in the search list for donors.

H. Donor Details Screen

The Donor Details screen provides full information about a selected blood donor. It displays the donor's name, age, blood group, address, phone number, and verification status. A Google Map is integrated to show the donor's exact location for easy navigation. The screen design is clean and simple, allowing seekers to quickly view and connect with the donor if needed.

I. Requester Details Screen

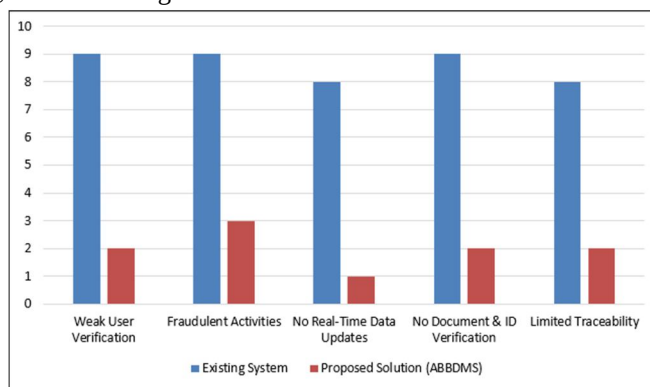
The Requester Details screen shows complete information about a selected blood requester. It includes the requester's name, required blood group, hospital name, address, reason for the request, and verification status. The screen also features an integrated Google Map showing the hospital location. The interface ensures that donors can easily verify the authenticity of a request before proceeding to help.

V. RESULT AND DISCUSSION

In this project, an extensive review of existing blood donation management systems was conducted to identify the current challenges and limitations. Many existing models focus primarily on basic registration and donor listing without ensuring the authenticity of users. Techniques such as simple database management and manual verification were observed, which are not sufficient to prevent fraudulent entries or fake blood requests. Furthermore, most systems lacked real-time updates, making emergency responses slower and less reliable.

Several research papers introduced methods like online donor tracking and basic location matching to connect donors and recipients. However, these systems did not effectively tackle the problem of verifying documents or ensuring the genuineness of the blood requests. Issues like data duplication, missing security features, and no integration of automated verification technologies were common drawbacks across the studied works. Additionally, the absence of strong notification and communication systems often led to delays in urgent cases.

Building upon these gaps, the proposed Blood Donation Management System introduces OTP-based phone number verification, OCR-based document verification for both donors and requesters, and real-time location mapping through Google Maps API. This multi-layered enhancement not only secures the platform but also ensures authenticity, reduces response times, and improves the reliability of the blood donation process. By addressing the critical flaws identified in the reviewed systems, the proposed solution offers a more efficient, transparent, and life-saving network.



The bar chart illustrates a comparison between the major drawbacks of the existing blood donor management systems and the improvements offered by the proposed Android-Based Blood Donor Management System (ABBDMS). In the existing system, key issues like weak user verification, high risk of fraudulent activities, lack of real-time updates, absence of proper document and identity verification, and limited traceability show high severity levels (rated between 8 and 9).

In contrast, the proposed ABBDMS addresses these challenges effectively by introducing OTP-based user verification, OCR-based fraud detection, real-time data synchronization through Firebase, secure document verification, and full donation traceability. These improvements significantly reduce the problem levels to a much lower range (1 to 3).

This visual comparison clearly highlights that ABBDMS not only minimizes the critical risks associated with the traditional system but also enhances user trust, system transparency, and operational efficiency, making it far more reliable for real-world emergency blood donation needs.

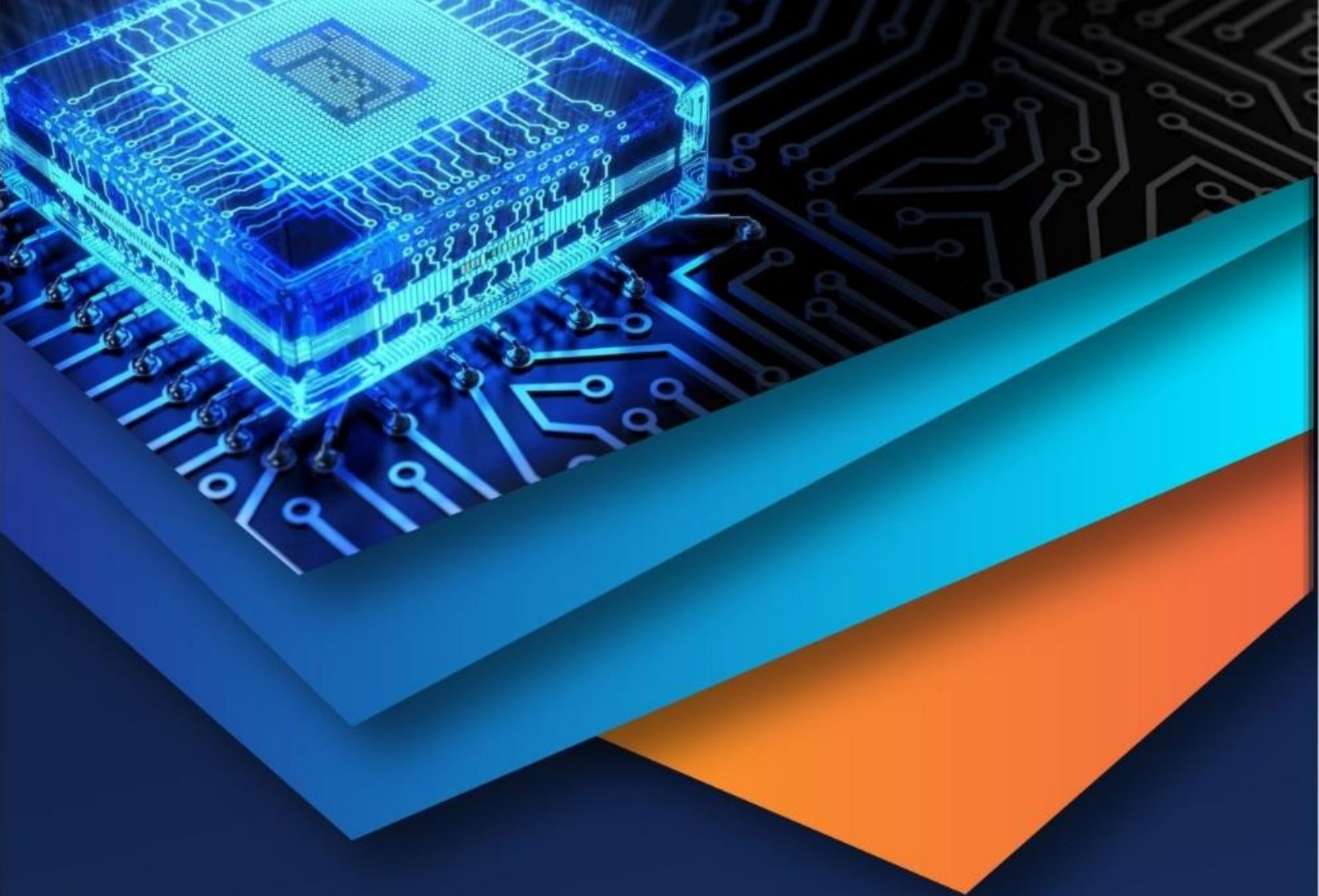
VI. CONCLUSION

The development of the Android-Based Blood Donor Management System (ABBDMS) marks a significant improvement over traditional systems by addressing core limitations such as weak authentication, static data handling, and poor traceability. Through the integration of OTP-based user verification, OCR-powered document validation, and real-time data synchronization with Firebase, the proposed system enhances security, accuracy, and user experience. The comparative analysis confirms that ABBDMS not only resolves existing issues but also establishes a scalable and efficient framework for future enhancements.

Looking ahead, the system provides a solid foundation for subsequent development phases involving UI/UX refinement, rigorous prototype testing, and real-world deployment. Future enhancements may include AI-powered donor matching, integration with hospital blood bank databases, handwritten ticket recognition, and real-time notification systems to further optimize blood donation processes and accessibility.

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